



रोल नं.



Roll No.

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कोड नं.
Q.P. Code

55/3/1

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-
पृष्ठ पर अवश्य लिखें।Candidates must write the Q.P. Code
on the title page of the answer-book.

भौतिक विज्ञान

PHYSICS (Theory)

निर्धारित समय: 2 घंटे

अधिकतम अंक : 50

Time Allowed : 2 hours

Maximum Marks : 50

नोट / NOTE

- (I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ 8 हैं।
Please check that this question paper contains 8 printed pages.
- (II) कृपया जाँच कर लें कि इस प्रश्न-पत्र में 18 प्रश्न हैं।
Please check that this question paper contains 18 questions.
- (III) प्रश्न-पत्र में दाहिने हाथ की एयर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख्य-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (IV) कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।
Please write down the serial number of the question in the answer-book at the given place before attempting it.
- (V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.



General Instructions:

Read the following instructions very carefully and strictly follow them :

- (i) This question paper contains **18** questions. **All** questions are **compulsory**.
- (ii) This Question paper is divided into **FIVE** sections- Sections **A, B, C, D** and **E**.
- (iii) In **Section A** : Question numbers **1 to 5** are Very Short Answer (VSA) type questions. Each question carries **1** mark.
- (iv) In **Section B**: Question numbers **6 to 9** are Very Short Answer (VSA) type questions. Each question carries **2** marks.
- (v) In **Section C**: Question numbers **10 to 13** are Short Answer (SA) type questions. Each question carries **3** marks.
- (vi) In **Section D** : Question number **14 and 15** are Case Study-Based questions carrying **4** marks each.
- (vii) In **Section E** : Question numbers **16 to 18** are Long Answer (LA) type questions. Each question carries **5** marks.
- (viii) There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the Sections except Section A.
- (ix) Kindly note that there is a separate question paper for Visually Impaired candidates.
- (x) Use of calculators is **NOT** allowed.

You may use the following values of physical constants wherever necessary :

$$c = 3 \times 10^8 \text{ m/s}$$

$$h = 6.63 \times 10^{-34} \text{ J s}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$$

$$\text{Mass of electron (me)} = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Mass of neutron} = 1.675 \times 10^{-27} \text{ kg}$$

$$\text{Mass of proton} = 1.673 \times 10^{-27} \text{ kg}$$

$$\text{Avogadro's number} = 6.023 \times 10^{23} \text{ per gram mole}$$

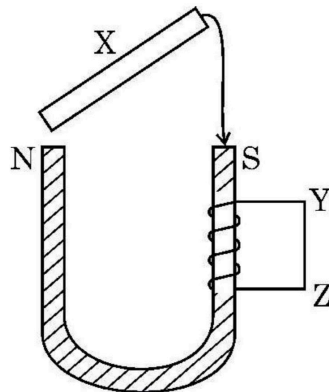
$$\text{Boltzmann's constant} = 1.38 \times 10^{-23} \text{ J K}^{-1}$$



SECTION - A

This section contains 5 Very Short Answer (VSA) Questions carrying 1 mark each.

1. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, then what will the ratio $\frac{v_A}{v_B}$ of the drift velocities of the electrons in them be? 1
2. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT}) \hat{j} + (0.6 \text{ mT}) \hat{k}$ is switched on in the region. What is the force acting on the segment? 1
3. You are required to design an air-filled solenoid of inductance 0.016 H having a length 0.81 m and radius 0.02 m. What should be the number of turns in the solenoid? 1
4. A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H . The area of the surface of the liquid through which light from the source can emerge out is? 1
5. A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet. 1



During fall of the rod, what will be the direction of current?

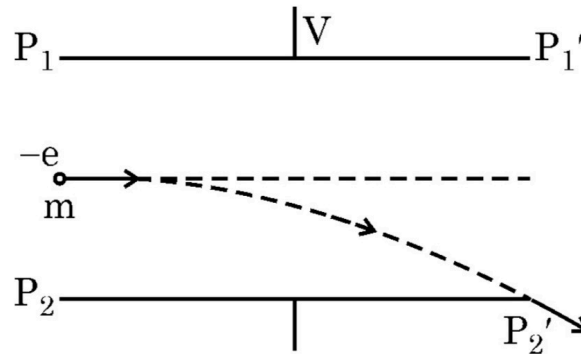
SECTION - B

This section contains 4 Very Short Answer (VSA) type questions carrying 2 marks each.

6. A light beam converges at a point O. In the path of this beam, a concave lens of focal length 15 cm is placed at a distance of 10 cm before point O. The beam now converges at a point O'. Find the magnitude and the direction of shift OO'. 2
7. n identical cells, each of e.m.f. E and internal resistance r , are connected in series. Later on, it was found out that two cells 'X' and 'Y' are connected in reverse polarities. Calculate the potential difference across the cell 'X'. 2



8. Figure shows a narrow beam of electrons entering with a velocity of 3×10^7 m/s, symmetrically through the space between two parallel horizontal plates P_1P_1' and P_2P_2' kept 2 cm apart.



2

If each plate is 3 cm long, calculate the potential difference V applied between the plates so that the beam just strikes the end P_2' .

9. A telescope has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate its magnifying power in normal adjustment and the distance of the image formed by the objective.

2

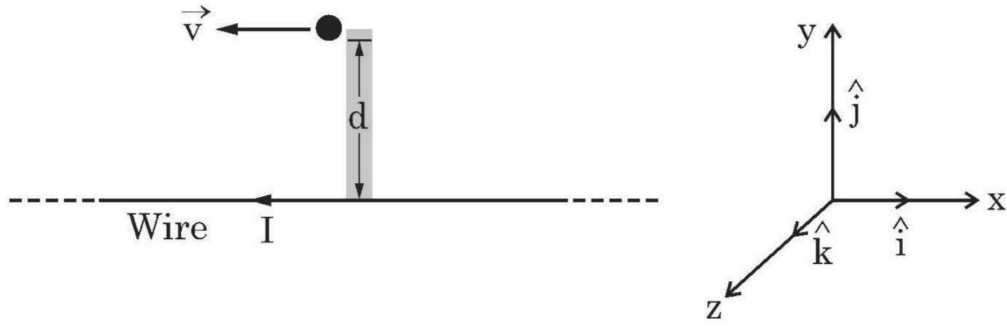
SECTION - C

This section contains 4 Short Answer (SA) type questions carrying 3 marks each.

10. (a) A conductor of length l is connected across an ideal cell of emf E . Keeping the cell connected, the length of the conductor is increased to $2l$ by gradually stretching it. If R and R' are initial and final values of resistance and v_d and v_d' are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) v_d' and v_d 3
- (b) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction?
11. (a) State Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify. 3
- (b) A metal rod of length 2 m is rotated with a frequency 60 rev/s about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2T perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. Induced between the centre and the end of the rod.



12. A particle of charge q is moving with a velocity $\vec{v}$ at a distance ' d ' from a long straight wire carrying a current ' I ' as shown in figure.

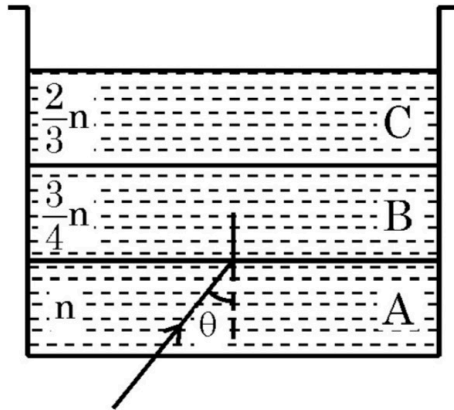


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At this instant, it is subjected to a uniform electric field $\vec{E}$ such that the particle keeps moving undeviated. In terms of unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$, find —

- the magnetic field $\vec{B}$
 - the magnetic force $\vec{F}_m$, and
 - the electric field $\vec{E}$ acting on the charge
13. (a) State two conditions under which total internal reflection occurs.
- (b)

3



A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n , $\frac{3}{4}n$ and $\frac{2}{3}n$, respectively. A laser beam is incident at the interface between A and B at angle θ as shown in figure. Prove that the beam does not enter region C at all for $\sin\theta \geq \frac{2}{3}$.

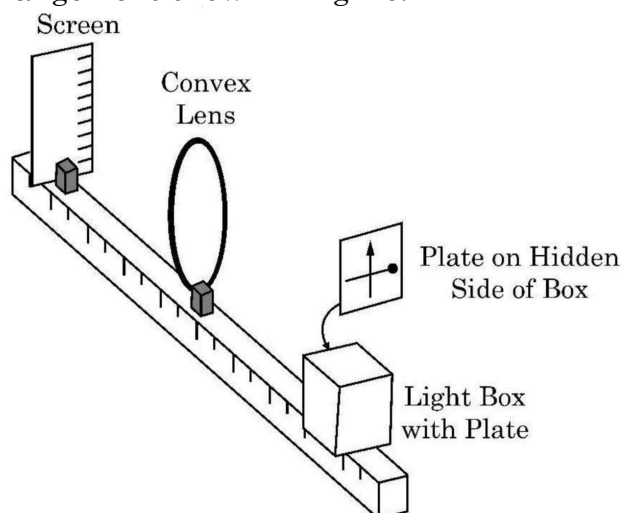


SECTION - D

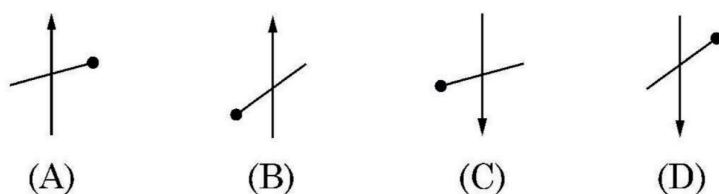
This section consists of 1 case-based/source-based questions carrying 4 marks each.

14. A thin lens is a transparent optical medium bounded by two surfaces, at least one of which should be spherical. Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, one can obtain the 'lens maker formula' and then the 'lens formula'. A lens has two foci-called 'first focal point' and 'second focal point' of the lens, one on each side. 4

(i) Consider the arrangement shown in figure.



A black vertical arrow and a horizontal thick line with a ball are painted on a glass plate. It serves as the object, when the plate is illuminated, its real image is formed on the screen. Which of the following correctly represents the image formed on the screen?



(ii) Which of the following statements is incorrect?

- (A) For a convex mirror, magnification is always negative.
- (B) For all virtual images formed by a mirror magnification is positive.
- (C) For a concave lens magnification is always positive.
- (D) For real and inverted images, magnification is always negative.

(iii) A convex lens of focal length ' f ' is cut into two equal parts perpendicular to the principal axis. The focal length of each part will be :

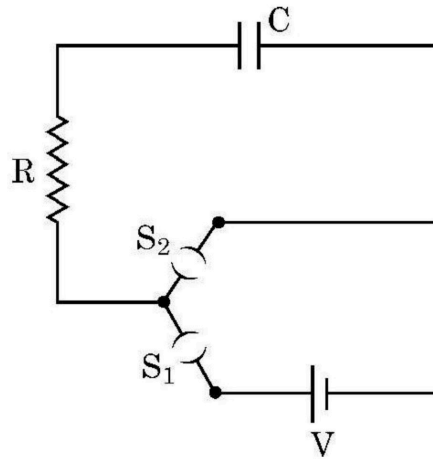
- (A) f
- (B) $2f$
- (C) $\frac{f}{2}$
- (D) $\frac{f}{4}$



- (iv) The distance of an object from the first focal point of a biconvex lens is X_1 and distance of the image from second focal point is X_2 . The focal length of the lens is :

- (A) $X_1 X_2$ (B) $\sqrt{X_1 + X_2}$
 (C) $\sqrt{X_1 X_2}$ (D) $\sqrt{\frac{X_2}{X_1}}$

15. A circuit consisting of a capacitor C , a resistor of resistance R and an ideal battery of emf V , as shown in figure is known as RC series circuit.



As soon as the circuit is completed by closing key S_1 (keeping S_2 open) charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference V_c ($= q/C$) across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged ($Q = VC$). During this process of charging, the charge q on the capacitor changes with time t as

$$q = Q[1 - e^{-t/RC}]$$

The charging current can be obtained by differentiating it and using

$$\frac{d}{dx}(e^{mx}) = me^{mx}$$

Consider the case when $R = 20 \text{ k}\Omega$, $C = 500 \text{ }\mu\text{F}$ and $V = 10 \text{ V}$.

- The final charge on the capacitor, when key S_1 is closed and S_2 is open, is
- For a sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor?
- What is the dimensional formula for RC ?
- The key S_1 is closed and S_2 is open. What is the value of current in the resistor after 5 seconds?



SECTION - E

This section consists of 3 Long Answer (LA) type questions carrying 5 marks each.

17. (a) (i) What is meant by current sensitivity of a galvanometer? Mention the factors on which it depends.
(ii) A galvanometer of resistance G is converted into a voltmeter of range $(0 - V)$ by using a resistance R . Find the resistance in terms of R and G , required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.
- (b) The magnetic flux through a coil of resistance 5 increases with time as :
 $\phi = (2t^3 + 5t^2 + 6t)$ mWb
Find the magnitude of induced current through the coil at $t = 2$ s.
17. (a) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. 5
Using this diagram, derive the mirror formula.
- (b) A concave mirror produces 2 times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.
18. (a) A small conducting sphere A of radius r charged to a potential V , is enclosed by a spherical conduction shell B of radius R . If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B. 5
- (b) What is the difference between an open surface and a closed surface? Draw elementary surface vector $d\vec{S}$ for a spherical surface S .
- (c) A small spherical shell S_1 has point charges $q_1 = -3 \mu\text{C}$, $q_2 = -2 \mu\text{C}$ and $q_3 = 9 \mu\text{C}$ inside it. This shell is enclosed by another big spherical shell S_2 . A point charge Q is placed in between the two surfaces S_1 and S_2 . If the electric flux through the surface S_2 is four times the flux through surface S_1 , find charge Q .

